

CYBERSECURITY INCIDENT RESPONSE DOCUMENTATION & REPORT

Microsoft Sentinel | Defender XDR | NIST SP 800-61 Rev.3 Aligned

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Document Control Notice

The document contains sensitive security information relating to an independent lab simulation. Distribution is restricted to authorised reviewers only. All activities described were conducted in an isolated Azure lab environment on personally own resources.

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1. Executive Summary

This document provides a formal incident response record for two high-severity security incidents detected during an enterprise SOC simulation project built on Microsoft Azure. Both incidents were generated through authorised adversary simulation using Atomic Red Team and were successfully detected by custom analytics rules in Microsoft Sentinel within approximately 30 - 38 minutes of execution.

1.1 Scope

This report covers the following domains:

- Security architecture and tool integration (Sentinel and Defender XDR)
- Incident detection and alert triage workflows
- NIST SP 800-61 Rev.3 four-phase incident response lifecycle
- MITRE ATT&CK mapped threat hunting procedures
- KQL-based detection rules and analytics
- Post incident review and lessons learned

1.2 Incident Overview

Field	Details
Incidents Covered	INC-5 (PowerShell Execution) INC-11 (Registry Persistence)
Total Incidents Generated	11 high-severity incidents across the lab session
Detection Platform	Microsoft Sentinel; Scheduled analytics rules
Affected Environment	Azure-hosted SOC lab; SOC-Workstation (10.0.1.10)
Attack Simulation Tool	Atomic Red Team (MITRE ATT&CK mapped)
Mean Time to Detect	Approximately 30 - 38 minutes depending on the incident
Mean Time to Respond	Approximately 10 - 11 minutes depending on the incident
Final Classification	Informational; Security Testing (Benign Positive)
Overall Outcome	Both incidents successfully detected, investigated, and closed

2. Security Environment Architecture

2.1 Architecture Overview

The lab environment was built entirely on Microsoft's cloud security tools. Hosted on Azure and designed to work together as one connected detection and response platform.

Components	Category	Function
Microsoft Sentinel	SIEM	Central log ingestion, analytics rule engine, incident management, playbook, threat hunting.
Log Analytics Workspace	Data Store	Backend data storage and KQL query engine for all Sentinel data.
Microsoft Defender XDR	Unified Portal	Unified SOC portal for incident investigation and analytics rule management
Azure Monitor Agent + DCR	Collection	Data Collection Rules with AMA for VM log ingestion and forwarding
Sysmon v15.20	Endpoint Telemetry	Rich process creation, network connection, registry, and file event logging
Atomic Red Team	Simulation	MITRE ATT&CK mapped adversary technique simulation framework

2.2 Data Ingestion & Log Sources

Log Source	Key Events
Windows Security Events via AMA + DCR	Logon (4624/4625), process (4688), privilege use
Sysmon via custom DCR	Process creation (EID 1), network connections (EID 3), registry writes (EID 13)
Linux Syslog via AMA + DCR	Auth, cron, and syslog facilities from Ubuntu 22.04 LTS endpoint

2.3 Analytics Rule Architecture

Four custom scheduled analytics rules were deployed in Microsoft Sentinel prior to any simulation activity. All rules run on a 5-minute schedule and are mapped to MITRE ATT&CK framework techniques.

Rule Name	MITRE Technique
Multiple Failed Logons: Possible Brute Force	T1110: Brute Force, Credential Access
Suspicious PowerShell Execution Detected	T1059.001: PowerShell, Execution
Suspicious Registry Run Key Persistence	T1547.001: Registry Run Keys, Persistence
RDP Brute Force Attack Detected	T1110: Brute Force via RDP, Credential Access

3. Phase 1: Preparation

All detection capability, tooling, and infrastructure was deployed and validated before any adversary simulation was conducted. This section documents exactly what was put in place in the environment at the time of the incidents.

3.1 Environment

- Azure-hosted SOC lab within Resource Group: SOC-Sentinel-Threat-Hunt-RG (UK South)
- Virtual Network: SOC-VNet (10.0.0.0/16) with SOC-Subnet (10.0.1.0/24)
- NSG configured with RDP and SSH access restricted to analyst home IP only

3.2 Endpoints

VM Name	Configuration
SOC-Workstation	Windows Server 2025 Datacenter 10.0.1.10 Sysmon v15.20 installed
SOC-Target-Win	Windows 11 Pro 10.0.1.20 Sysmon v15.20 installed
SOC-Linux	Ubuntu 22.04 LTS 10.0.1.30 auditd and rsyslog configured

3.3 Detection Capability

- Microsoft Sentinel SIEM deployed on Log Analytics Workspace (SOC-Sentinel-Threat-Hunt-LAW)
- Sysmon v15.20 installed on all Windows endpoints using SwiftOnSecurity configuration, providing process creation, network connection, registry modification, and file creation telemetry
- Azure Monitor Agent configured with two Data Collection Rules: SOC-Windows-DCR for Windows Security Events and SOC-Sysmon-DCR for Sysmon telemetry
- Four custom scheduled analytics rules created and enabled prior to any simulation activity, all mapped to MITRE ATT&CK techniques
- Log flow validated end to end via KQL query confirming SecurityEvent data ingestion from both Windows endpoints

3.4 Incident Severity Classification and SLA Matrix

Priority	Definition and SLA
P1: Critical	Active data exfiltration, ransomware detonation, or domain compromise. MTDD < 5 min. MTTR < 30 min.
P2: High	Confirmed malicious execution, active persistence, or credential access. MTDD < 10 min. MTTR < 1 hour.
P3: Medium	Suspicious activity requiring investigation. MTDD < 30 min. MTTR < 4 hours.
P4: Low	Informational or policy violations. MTDD < 2 hours. MTTR < 24 hours.

Note: Both INC-5 and INC-11 were classified as P2; High based on the nature of the detected techniques (code execution and persistence). Both met SLA targets; MTTD approximately 10 minutes, MTTR approximately 45 minutes.

4. Phase 2: Detection and Analysis

4.1 Alert Triage Workflow

The following triage workflow was applied to both incidents upon detection in Microsoft Defender XDR:

- Opened Microsoft Defender XDR Incidents queue filtered by Status: New/Active and Severity: High
- Selected incident; reviewed title, severity, MITRE tactic tag, alert count, and creation timestamp
- Read incident description and timeline, identified triggering analytics rule and correlated Sysmon events
- Reviewed raw query results in Alerts tab; confirmed Sysmon process creation and registry write events as primary evidence
- Classified incident; determined Benign Positive based on confirmed authorised Atomic Red Team simulation activity
- Assigned incident to analyst account, added investigation comment, updated status to Active
- Closed incident with Benign Positive classification and documented rationale

4.2 Incident Details: INC-5: Suspicious PowerShell Execution Detected

Incident Details	Value
Incident ID	INC-5
Incident Title	Suspicious PowerShell Execution Detected
Severity / Priority	High - P2
Date & Time Detected	May 12, 2026, 18:19:34 PM UTC+1
First Activity	May 12, 2026, 17:49:51 PM UTC+1
Last Activity	May 12, 2026, 18:13:59 PM UTC+1
Date & Time Resolved	May 12, 2026, 18:30 PM UTC+1
MTTD	Approximately 30 minutes
MTTR	Approximately 11 minutes
Detection Source	Microsoft Sentinel; Scheduled Analytics Rule
Affected Host	SOC-Workstation 10.0.1.10 Windows Server 2025
User Context	socadmin (local administrator)
Attack Vector	Encoded PowerShell execution via Atomic Red Team T1059.001

Kill Chain Stage	Execution phase. No lateral movement or C2 established
MITRE ATT&CK TTPs	(T1059; T1059.001) - Command and Scripting Interpreter: PowerShell
Analyst Assigned to	Jennifer Hope - SOC Analyst
Final Status	RESOLVED: Post-Incident Review Complete
Classification	Informational: Security Testing (Benign Positive)

Detection Method: INC-5

Alert triggered by custom Sentinel analytics rule 'Suspicious PowerShell Execution Detected'; a scheduled KQL query running every 5 minutes monitoring Sysmon Event ID 1 (process creation) for PowerShell execution containing suspicious execution flags including -nop (no profile), -w hidden (hidden window), -enc (encoded command), and bypass (execution policy bypass). These four flags in combination represent the standard living-off-the-land PowerShell pattern used in the majority of real ransomware and APT intrusions.

KQL Query for INC-5

```
1  Event
2  | where Source == "Microsoft-Windows-Sysmon"
3  | where EventID == 1
4  | where RenderedDescription contains "powershell"
5  |   and (RenderedDescription contains "-enc"
6  |         or RenderedDescription contains "-nop"
7  |         or RenderedDescription contains "bypass"
8  |         or RenderedDescription contains "hidden")
9  | project TimeGenerated, Computer, RenderedDescription
```

Evidence: INCE-5

8 correlated Sysmon process creation events identified in query results, all originating from SOC-Workstation. Events confirmed PowerShell execution with encoded command parameters consistent with attacker tradecraft. The Atomic Red Team T1059.001 test executed Mimikatz v2.2.0 via PowerShell, which attempted sekurlsa::logonpasswords to extract password hashes from Windows memory. Sysmon Event ID 1 captured the full execution chain including the process command line arguments.

Note: Entity mapping was not configured in the analytics rule KQL. As a result, the Assets tab in Defender XDR showed 0 devices and 0 users. Entity information was obtained manually from the raw query results in the Alerts tab. This is documented as a lesson learned. See Section 7.

4.3 Incident Details: INC-11: Suspicious Registry Run Key Persistence

Incident Details	Value
Incident ID	INC-11
Incident Title	Suspicious Registry Run Key Persistence
Severity / Priority	High - P2

Date & Time Detected	May 12, 2026, 18:31:41 PM UTC+1
First Activity	May 12, 2026: 17:53:41 PM UTC+1
Last Activity	May 12, 2026: 18:14:24 PM UTC+1
Date & Time Resolved	May 12, 2026, 18:41 PM UTC+1
MTTD	Approximately 38 minutes
MTTR	Approximately 10 minutes
Detection Source	Microsoft Sentinel: Scheduled Analytics Rule
Affected Host	SOC-Workstation 10.0.1.10 Windows Server 2025
User context	socadmin (local administrator)
Attack Vector	Registry Run Key write via Atomic Red Team T1547.001
Kill Chain Stage Reached	Persistence phase; execution confirmed, no lateral movement
MITRE ATT&CK TTPs	T1547.001 - Boot or Logon Autostart Execution: Registry Run Keys
Analyst Assigned to	Jennifer Hope - SOC Analyst
Final Status	RESOLVED: Post-Incident Review Complete
Classification	Informational: Security Testing (Benign Positive)

Detection Method: INC-11

Alert triggered by custom Sentinel analytics rule 'Suspicious Registry Run Key Persistence'; a scheduled KQL query running every 5 minutes monitoring Sysmon Event ID 13 (registry value set) for write operations targeting the CurrentVersion\Run registry key path. This key path is used by Windows to execute programs automatically at user logon and is one of the most abused persistence mechanisms by ransomware and malware families including WannaCry and LockBit.

KQL Query for INC-11

```

1  Event
2  | where Source == "Microsoft-Windows-Sysmon"
3  | where EventID == 13
4  | where RenderedDescription contains "CurrentVersion\\Run"
5  | project TimeGenerated, Computer, RenderedDescription

```

Evidence: INC-11

Sysmon Event ID 13 (registry value set) confirmed a value named SOCTestPersistence was written to HKCU\Software\Microsoft\Windows\CurrentVersion\Run with data pointing to C:\Windows\System32\calc.exe. This was the direct output of the Atomic Red Team T1547.001 test

which completed with exit code 0 (success). The registry write was captured by Sysmon and forwarded to Sentinel via the SOC-Sysmon-DCR Data Collection Rule.

Note: MTTR reflects time to close a confirmed Benign Positive where full context was already known to the analyst. In a real unknown incident MTTR would be significantly longer.

5. Phase 3: Containment, Eradication and Recovery

5.1 Containment

Activity was confirmed as authorised security testing. No containment actions were required in the lab environment. The following steps document what containment would involve in a production environment:

- Isolate the affected endpoint from the network via Microsoft Defender for Endpoint device isolation. This cuts all network connections while preserving the ability to investigate via Live Response
- Block the offending process hash at the EDR level: This prevents the same executable from running on any other MDE-enrolled device in the tenant. A process hash is a unique fingerprint of the executable file; blocking it means Defender will terminate any future execution of that exact file automatically
- Revoke the socadmin credentials and force an immediate password reset to prevent lateral movement using compromised credentials
- Preserve forensic artifacts before any remediation, collect memory dump and process list via MDE Live Response

5.2 Eradication

In a production environment, eradication steps would include:

- Scan for persistence mechanisms using Sysinternals Autoruns and Sysmon Event ID 13 registry monitoring to identify all run key entries
- Review all scheduled tasks, startup folders, and registry run keys for unauthorised entries across all endpoints
- Validate no lateral movement occurred via Event ID 4624 and 4625 logon analysis across all endpoints in the environment
- Search for additional Mimikatz indicators; Event ID 10 (process access) targeting lsass.exe and any suspicious LSASS memory reads

5.3 Recovery

In a production environment, recovery steps would include:

- Restore affected systems from a known clean snapshot taken prior to the incident
- Re-enable full monitoring and confirm baseline telemetry is flowing correctly before returning the system to production
- Validate no reinfection within a 72-hour observation period before closing the incident fully
- Confirm all IOCs have been blocked across the environment before restoring network access

5.4 Project Outcome

Both incidents confirmed as authorised Atomic Red Team simulation. T1059.001 detected via Sysmon process creation telemetry. T1547.001 detected via Sysmon registry modification telemetry. Both analytics rules performed as designed. No remediation required. Incidents closed as Benign Positive.

6. Phase 4: Post-Incident Activity

6.1 Post-Incident Review (PIR)

PIR Element	Findings for This Project
Timeline Reconstruction	First simulation commands executed at approximately 17:49 UTC+1. First alert fired at approximately 18:19 UTC+1. Incident investigated and closed by 18:30 UTC+1. Total elapsed time from simulation to closure: approximately 41 minutes.
Root Cause	Authorised adversary simulation using Atomic Red Team. T1059.001 executed via encoded PowerShell. T1547.001 executed via registry write. Both techniques are standard attacker tradecraft mapped to MITRE ATT&CK.
Kill Chain Mapping	Both incidents contained at the Execution and Persistence phases. No lateral movement, credential theft confirmation, or C2 communication was established. Defence succeeded at initial detection.
ATT&CK TTP Analysis	T1059.001, T1547.001, T1110.001; All three simulated techniques were detected by corresponding analytics rules. T1003 (Mimikatz credential dumping) was simulated but no dedicated rule existed for it. Documented as a detection gap.
Detection Gap Analysis	Entity mapping absent from all four analytics rules. MDE not onboarded; no EDR-level endpoint telemetry. No dedicated rule for T1003 credential dumping despite Mimikatz simulation.
Response Effectiveness	Both P2 SLA targets met. INC-5: MTTD 30 min, MTTR 11 min. INC-11: MTTD 38 min, MTTR 10 min. Triage workflow followed correctly. Incident documentation complete.
Business Impact	No business impact as it was carried out in an isolated lab environment. No production systems, real credentials, or sensitive data involved.

7. Lessons Learned and Recommendations

7.1 What worked well:

- All four custom analytics rule fired just as expected once an incident / threat was detected.
- Sysmon v15.20 with SwiftOnSecurity configuration provided sufficient process creation and registry modification telemetry to detect both PowerShell and persistence techniques
- MITRE ATT&CK mapping correctly identified the techniques and tactics of the incidents.
- End-to-end detection pipeline validated: VM to Sysmon to Azure Monitor Agent to Log Analytics to Sentinel to Incident in under 10 minutes
- NIST 800-61 four-phase framework applied correctly throughout the investigation and documentation process

7.2 Areas for improvement

- Entity mapping was not configured in analytics rule KQL. As a result, all four incidents showed 0 devices and 0 users in the Assets tab in Defender XDR. Future rules must include explicit extend statements to extract and map hostname and account entities so incident investigations have full context from the moment an alert fires.
- Microsoft Defender for Endpoint was not onboarded due to trial licensing limitations on a personal Azure pay-as-you-go account. Full MDE integration would provide richer endpoint telemetry, automated response capability, and correct entity population on incidents
- Multiple duplicate incidents were generated for the same technique across consecutive 5-minute rule evaluation windows. A production environment requires incident grouping configuration to deduplicate alerts from the same rule into a single incident over a defined time window
- No dedicated detection rule existed for T1003 credential dumping despite Mimikatz being executed via the T1059.001 Atomic Red Team test. Mimikatz targets the LSASS process which is detectable via Sysmon Event ID 10 (process access). A dedicated rule should be created

7.3 Recommendations

Recommendation	Priority
Add specific entity mapping to all KQL analytics rules using extend statements to extract hostname and account fields	High
Onboard endpoints to Microsoft Defender for Endpoint for EDR-level visibility and automated response capability	High
Configure incident grouping rules in Sentinel to deduplicate alerts from the same analytics rule within a defined time window	Medium
Create a dedicated T1003 credential dumping detection rule targeting Sysmon Event ID 10 LSASS process access	Medium
Create automation playbooks in Sentinel to automatically assign and tag incidents on creation, reducing manual triage overhead	Medium
Implement a formal alert tuning process, review duplicate incidents weekly and refine rule thresholds accordingly	Low

Appendix A: Screenshots

The following screenshots are evidence of this incident response report. All screenshots were captured during the live lab session.

All screenshots referenced below are available in the project repository at: [GitHub](#)

Screenshot	Purpose
Azure Resource Group: SOC-Sentinel-Threat-Hunt-RG overview	Confirms lab environment and resource organisation
Virtual Network and NSG configuration: SOC-Subnet 10.0.1.0/24	Confirms network architecture and security controls
Sysmon installation confirmation: Sysmon64 started on both Windows VMs	Confirms endpoint telemetry operational
Log Analytics Workspace and Sentinel deployment	Confirms SIEM platform operational
Data Collection Rules: SOC-Windows-DCR and SOC-Sysmon-DCR	Confirms log ingestion pipeline configured
KQL query results: SecurityEvent data flowing from both VMs	Confirms end-to-end log pipeline validated
Four custom analytics rules showing Enabled status	Confirms detection capability in place before simulation
Atomic Red Team installation and three test executions	Confirms authorised simulation activity
Incidents queue: 11 high-severity incidents generated	Confirms detection pipeline produced results
INC-5: Attack Story tab showing event timeline	Primary investigation evidence for PowerShell incident
INC-5: Alerts tab showing 8 correlated Sysmon events	Forensic evidence of PowerShell execution
INC-5: Managed incident with classification and analyst comment	Incident handling and closure documentation
INC-11: Incidents queue entry showing Registry Persistence	Detection evidence for persistence incident

Appendix B: MITRE ATT&CK Techniques Simulated

Technique ID and Name	How Simulated and Detected
T1059.001: PowerShell	Atomic Red Team test executed Mimikatz via encoded PowerShell. Detected by Sysmon EID 1 process creation rule monitoring suspicious PowerShell flags.
T1547.001: Registry Run Keys	Atomic Red Team test wrote SOCTestPersistence value to HKCU CurrentVersion Run. Detected by Sysmon EID 13 registry modification rule.
T1110.001: SMB Brute Force	Manual simulation of 10 failed logon attempts. Detected by Windows Security EID 4625 aggregation rule.
T1110: RDP Brute Force	Manual failed logon simulation with LogonType 10. Detected by RDP brute force rule.
T1003: Credential Dumping	Mimikatz sekurlsa::logonpasswords executed via T1059.001 test. No dedicated detection rule. Identified as gap.

Appendix C: Tools Used

Tool	Purpose in This Project
Microsoft Azure (pay-as-you-go)	Cloud infrastructure: All VMs, networking, and security services
Microsoft Sentinel	SIEM: log storage, analytics rules, incident management, KQL hunting
Microsoft Defender XDR	Unified SOC portal: incident investigation and analytics management
Log Analytics Workspace	Log storage database and KQL query engine
Azure Monitor Agent + DCR	Log collection and forwarding from endpoints to Sentinel
Sysmon v15.20	Endpoint telemetry: process, network, registry, and file event logging
SwiftOnSecurity Sysmon Config	A ready-made Sysmon configuration file built and maintained by the security community that tells Sysmon exactly what activity to monitor and log on each endpoint.
KQL (Kusto Query Language)	Detection logic, threat hunting queries, and analytics rules
Atomic Red Team	MITRE ATT&CK mapped adversary simulation framework

MITRE ATT&CK v15	Threat framework used for technique mapping and detection validation
NIST SP 800-61 Rev.3	Incident response framework used for documentation and methodology